

Test Commands

Windows Detection Engineering Lab — Wazuh, Sysmon & MITRE ATT&CK

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Project: Windows Detection Engineering Lab — Wazuh, Sysmon & MITRE ATT&CK
Detection Scope: 5 validated detections
Environment: Local VirtualBox SOC lab
Tools Used: Wazuh SIEM/XDR, Windows 10, Sysmon, Windows Security Logs, Windows Defender, Atomic Red Team, MITRE ATT&CK

1. Purpose of This Document

The purpose of this document is to list and explain the controlled commands used to generate telemetry for the Windows Detection Engineering Lab. Each command was executed inside a local Windows 10 lab VM to safely trigger endpoint activity that could be collected by Wazuh and validated through custom detection rules.

This document supports the detection validation process by showing exactly how each alert was generated, which telemetry source was expected to capture the activity, what custom Wazuh rule was expected to fire, and what evidence should be reviewed after execution.

The commands in this document were used only for controlled cybersecurity learning, SIEM validation, and detection engineering documentation. No real malware, credential theft, ransomware, LSASS dumping, unauthorized scanning, command-and-control activity, or third-party system testing was performed.

2. Validated Detection Scope

The final validated scope of this project includes five detections:

Detection ID	Detection Name	Custom Rule ID	MITRE ATT&CK Technique	Telemetry Source
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DET-001	PowerShell Encoded Command Execution	100100	T1059.001	Sysmon / Windows Process Telemetry
DET-002	Local User Account Creation	100101	T1136.001	Windows Security Logs
DET-003	Windows Service Creation / Modification	100102	T1543.003	Windows Service / Process Telemetry
DET-009	EICAR Malware-Test File Detection	100108	T1204.002	Windows Defender Operational Logs
DET-010	Atomic Red Team Framework Activity	100109	T1059.001	Sysmon Event ID 1

Only these detections are included because they were tested and supported with Wazuh dashboard evidence.

3. Lab Execution Rules

All commands in this document must be executed only inside the authorized lab environment.

The intended execution environment is:

Component	Role
Windows 10 VM	Endpoint where test commands are executed
Wazuh Agent	Collects Windows telemetry and forwards it to Wazuh
Wazuh Manager	Processes logs and applies detection rules
Wazuh Dashboard	Used to validate alerts and collect evidence
Sysmon	Provides process and command-line telemetry
Windows Defender	Provides EICAR detection telemetry

Before running the commands, the following conditions should be confirmed:

1. The Windows 10 VM is powered on.
2. The Wazuh agent is installed and running.
3. Sysmon is installed and running.
4. The Windows endpoint appears as active in Wazuh Dashboard.

5. The relevant Windows event channels are configured for collection.
 6. Custom Wazuh rules are loaded without XML errors.
 7. The lab is isolated and authorized for testing.
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4. General Validation Workflow

For each detection, the same validation workflow was used:

1. Confirm the Windows endpoint is connected to Wazuh.
2. Execute the controlled test command on the Windows 10 VM.
3. Wait for the event to be collected and processed by Wazuh.
4. Open Wazuh Dashboard.
5. Search for the expected custom rule ID.
6. Review alert details and JSON fields.
7. Capture screenshot evidence.
8. Document whether the detection fired successfully.
9. Perform cleanup where required.
10. Record false-positive and triage notes.

This process ensures that detections are not only written, but actually validated.

DET-001: PowerShell Encoded Command Execution

Detection Objective

The objective of DET-001 is to generate PowerShell encoded command telemetry and validate that Wazuh can detect encoded PowerShell execution. Encoded PowerShell commands are often used to hide command content from casual review. In this lab, the encoded command is harmless and only runs `whoami`.

Field	Value
Detection ID	DET-001
Detection Name	PowerShell Encoded Command Execution

Custom Rule ID	100100
MITRE ATT&CK Technique	T1059.001
Tactic	Execution
Expected Telemetry	PowerShell process execution with encoded command-line argument
Execution Location	Windows 10 VM
Required Privileges	Standard user or Administrator

Test Command

Run the following command in **Windows PowerShell** on the Windows 10 VM:

```
$Command = 'whoami'  
$Bytes = [System.Text.Encoding]::Unicode.GetBytes($Command)  
$EncodedCommand = [Convert]::ToBase64String($Bytes)  
powershell.exe -NoProfile -EncodedCommand $EncodedCommand
```

What the Command Does

This command performs the following actions:

Step	Action
1	Stores the harmless command whoami inside a PowerShell variable
2	Converts the command into Unicode bytes
3	Encodes the command into Base64
4	Executes PowerShell with the -EncodedCommand parameter
5	Generates process telemetry for Wazuh and Sysmon to collect

The command is safe because it only runs `whoami`, which displays the current user context. The purpose is to generate telemetry that resembles encoded PowerShell behavior without executing harmful actions.

Expected Telemetry

The expected telemetry should show:

Expected Field	Expected Value / Pattern
Process	<code>powershell.exe</code>
Command Line	Contains <code>-EncodedCommand</code>
User	Lab user account
Parent Process	PowerShell, terminal, explorer, or command shell
MITRE Technique	T1059.001
Wazuh Custom Rule	100100

Wazuh Dashboard Search

After running the command, go to Wazuh Dashboard and search:

`rule.id:100100`

If the custom rule does not immediately appear, search for:

`powershell`

or:

`EncodedCommand`

or:

`T1059.001`

Expected Detection Result

The expected result is:

Item	Expected Result
Wazuh Alert	Generated
Custom Rule	100100
Detection Status	Detected
MITRE Mapping	T1059.001
Evidence Screenshot	Captured from Wazuh Dashboard

Evidence to Capture

The following evidence should be captured:

1. Screenshot of the PowerShell command execution.
2. Wazuh alert screenshot showing rule **100100**.
3. JSON view screenshot showing command-line details.
4. MITRE ATT&CK mapping screenshot if available.

Recommended evidence filename:

04-powershell-alert.png

Recommended sanitized JSON filename:

powershell-encoded-command-alert.json

Cleanup Required

No cleanup is required for DET-001 because the command does not create files, users, services, or persistent changes.

Safety Notes

This command does not execute malware or download external content. It is used only to generate encoded PowerShell telemetry for detection validation.

DET-002: Local User Account Creation

Detection Objective

The objective of DET-002 is to generate local Windows account creation telemetry and validate that Wazuh can detect account creation activity. Local account creation can be used legitimately by administrators, but it may also indicate persistence if performed by an attacker.

Field	Value
Detection ID	DET-002
Detection Name	Local User Account Creation
Custom Rule ID	100101
MITRE ATT&CK Technique	T1136.001
Tactic	Persistence
Expected Telemetry	Local user account creation event
Execution Location	Windows 10 VM
Required Privileges	Administrator

Test Command

Run the following command in **PowerShell as Administrator** on the Windows 10 VM:

```
New-LocalUser -Name "lab_t1136_test" -NoPassword
```

Verification Command

After creating the user, verify that the account exists:

```
Get-LocalUser | Where-Object {$_.Name -eq "lab_t1136_test"}
```

Cleanup Command

After evidence has been collected, remove the test account:

```
Remove-LocalUser -Name "lab_t1136_test"
```

What the Command Does

This command performs the following actions:

Step	Action
1	Creates a temporary local user named <code>lab_t1136_test</code>
2	Generates Windows account management telemetry
3	Allows Wazuh to process the event
4	Triggers the custom local account creation detection
5	Removes the account after validation

The account is created only for lab validation and should be removed immediately after the alert is captured.

Expected Telemetry

The expected telemetry should show:

Expected Field	Expected Value / Pattern
Event Type	Local user account creation
Created Account	<code>lab_t1136_test</code>

Creator Account	Lab administrator account
Log Source	Windows Security logs
MITRE Technique	T1136.001
Wazuh Custom Rule	100101

Wazuh Dashboard Search

After running the command, go to Wazuh Dashboard and search:

rule.id:100101

If the custom rule does not appear immediately, search for:

lab_t1136_test

or:

T1136.001

or:

user account created

Expected Detection Result

The expected result is:

Item	Expected Result
Wazuh Alert	Generated
Custom Rule	100101
Detection Status	Detected
MITRE Mapping	T1136.001
Evidence Screenshot	Captured from Wazuh Dashboard

Cleanup

Test account removed

Evidence to Capture

The following evidence should be captured:

1. Screenshot of the user creation command.
2. Wazuh alert screenshot showing rule **100101**.
3. JSON view screenshot showing account creation fields.
4. MITRE ATT&CK mapping screenshot if available.
5. Optional screenshot showing the test account was removed.

Recommended evidence filename:

05-local-user-created-alert.png

Recommended sanitized JSON filename:

local-user-created-alert.json

Cleanup Required

Cleanup is required for this detection.

Run:

```
Remove-LocalUser -Name "lab_t1136_test"
```

Confirm the account was removed:

```
Get-LocalUser | Where-Object {$_.Name -eq "lab_t1136_test"}
```

If no output appears, the account has been removed successfully.

Safety Notes

This test creates a temporary local user account only inside the lab VM. The account should not be used for login or privilege escalation. It should be removed after detection validation.

DET-003: Windows Service Creation / Modification

Detection Objective

The objective of DET-003 is to generate Windows service creation telemetry and validate that Wazuh can detect service creation or modification behavior. Windows service creation is commonly used for legitimate software installation, but it may also be abused by attackers for persistence or privilege escalation.

Field	Value
Detection ID	DET-003
Detection Name	Windows Service Creation / Modification
Custom Rule ID	100102
MITRE ATT&CK Technique	T1543.003
Tactic	Persistence / Privilege Escalation
Expected Telemetry	Windows service creation event
Execution Location	Windows 10 VM
Required Privileges	Administrator

Test Command

Run the following command in **Command Prompt as Administrator** on the Windows 10 VM:

```
sc.exe create WazuhLabSvc binPath= "C:\Windows\System32\cmd.exe /c whoami" start=demand
```

Verification Command

Check whether the test service exists:

```
sc.exe query WazuhLabSvc
```

Cleanup Command

After evidence has been collected, delete the test service:

```
sc.exe delete WazuhLabSvc
```

What the Command Does

This command performs the following actions:

Step	Action
1	Creates a temporary Windows service named <code>WazuhLabSvc</code>
2	Sets the service binary path to run a harmless <code>whoami</code> command
3	Generates service creation telemetry
4	Allows Wazuh to process the event
5	Triggers the custom service creation detection
6	Deletes the test service after validation

The service is created only to generate telemetry. It does not install malware and should be removed after validation.

Expected Telemetry

The expected telemetry should show:

Expected Field	Expected Value / Pattern
Service Name	<code>WazuhLabSvc</code>
Service Binary Path	<code>cmd.exe /c whoami</code>

Event Type	Service creation or modification
MITRE Technique	T1543.003
Wazuh Custom Rule	100102

Wazuh Dashboard Search

After running the command, go to Wazuh Dashboard and search:

rule.id:100102

If the custom rule does not appear immediately, search for:

WazuhLabSvc

or:

sc.exe

or:

T1543.003

Expected Detection Result

The expected result is:

Item	Expected Result
Wazuh Alert	Generated
Custom Rule	100102
Detection Status	Detected
MITRE Mapping	T1543.003
Evidence Screenshot	Captured from Wazuh Dashboard
Cleanup	Test service deleted

Evidence to Capture

The following evidence should be captured:

1. Screenshot of the service creation command.
2. Wazuh alert screenshot showing rule **100102**.
3. JSON view screenshot showing service name or command fields.
4. MITRE ATT&CK mapping screenshot if available.
5. Optional screenshot showing the service was deleted.

Recommended evidence filename:

06-service-creation-alert.png

Recommended sanitized JSON filename:

service-creation-alert.json

Cleanup Required

Cleanup is required for this detection.

Run:

```
sc.exe delete WazuhLabSvc
```

Confirm deletion:

```
sc.exe query WazuhLabSvc
```

If the service does not exist, Windows should return a service-not-found style message.

Safety Notes

This command creates a temporary test service inside the Windows lab VM. The service should not be started unless required for testing. It should be removed after telemetry is collected.

DET-009: EICAR Malware-Test File Detection

Detection Objective

The objective of DET-009 is to validate that Windows Defender malware-test telemetry is forwarded into Wazuh and detected by a custom rule. The EICAR test file is used because it is a safe anti-malware test file designed to trigger antivirus detection without using real malware.

Field	Value
Detection ID	DET-009
Detection Name	EICAR Malware-Test File Detection
Custom Rule ID	100108
MITRE ATT&CK Technique	T1204.002
Tactic	Execution
Expected Telemetry	Windows Defender detection/remediation event
Execution Location	Windows 10 VM
Required Privileges	Standard user or Administrator

Test Command

Run the following command in **PowerShell as Administrator** on the Windows 10 VM:

```
mkdir C:\Users\Public\WazuhMalwareSim
Set-Content -Path "C:\Users\Public\WazuhMalwareSim\eicar.com.txt" -Value
'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*'
```

What the Command Does

This command performs the following actions:

Step	Action
1	Creates a lab directory under <code>C:\Users\Public\WazuhMalwareSim</code>
2	Writes the EICAR anti-malware test string to a text file
3	Triggers Windows Defender detection or remediation
4	Generates Defender Operational log telemetry
5	Allows Wazuh to collect the Defender event
6	Triggers the custom EICAR detection rule

This command does not execute real malware. It uses a test string designed for antivirus validation.

Expected Telemetry

The expected telemetry should show:

Expected Field	Expected Value / Pattern
Event Source	Windows Defender Operational
Threat Name	EICAR or EICAR-Test-File
File Path	<code>C:\Users\Public\WazuhMalwareSim\eicar.com.txt</code>
Defender Action	Detected, quarantined, removed, or remediated
MITRE Technique	T1204.002
Wazuh Custom Rule	100108

Wazuh Dashboard Search

After running the command, go to Wazuh Dashboard and search:

rule.id:100108

If the custom rule does not appear immediately, search for:

EICAR

or:

Windows Defender

or:

Microsoft-Windows-Windows Defender/Operational

or:

T1204.002

Expected Detection Result

The expected result is:

Item	Expected Result
Windows Defender	Detects or remediates EICAR
Wazuh Alert	Generated
Custom Rule	100108
Detection Status	Detected
MITRE Mapping	T1204.002
Evidence Screenshot	Captured from Wazuh Dashboard

Evidence to Capture

The following evidence should be captured:

1. Screenshot of the EICAR command or Windows Defender detection.
2. Wazuh alert screenshot showing rule **100108**.
3. JSON view screenshot showing Defender event details.

4. File path or threat name evidence.
5. MITRE ATT&CK mapping screenshot if available.

Recommended evidence filenames:

13-eicar-defender-alert.png

13-eicar-defender-json.png

Cleanup Required

Windows Defender usually quarantines or removes the EICAR test file automatically. If the folder remains, it can be removed manually after evidence collection.

Run:

```
Remove-Item "C:\Users\Public\WazuhMalwareSim" -Recurse -Force
```

If Defender already removed the file, the command may show that the item does not exist. That is acceptable.

Safety Notes

EICAR is not real malware. It is used only to validate antivirus detection and telemetry forwarding. This test should still be performed only in the lab environment.

DET-010: Atomic Red Team Framework Activity

Detection Objective

The objective of DET-010 is to generate Atomic Red Team framework telemetry and validate that Wazuh can detect adversary-emulation framework activity through Sysmon process creation logs.

Atomic Red Team is a legitimate testing framework used by blue teams and detection engineers to validate security controls against MITRE ATT&CK techniques. In this project, it was used in a controlled and safe manner.

Field	Value
Detection ID	DET-010
Detection Name	Atomic Red Team Framework Activity
Custom Rule ID	100109
MITRE ATT&CK Technique	T1059.001
Tactic	Execution
Expected Telemetry	PowerShell process execution containing Atomic Red Team command activity
Execution Location	Windows 10 VM
Required Privileges	Administrator recommended

Installation Command

Run the following command in **PowerShell as Administrator** on the Windows 10 VM:

```
Install-Module -Name invoke-atomicredteam,powershell-yaml -Scope CurrentUser
```

If PowerShell asks to install the NuGet provider, type:

Y

If PowerShell asks whether to trust the repository, type:

Y

Import Module Command

After installation, import the module:

```
Import-Module Invoke-AtomicRedTeam
```

Safe Test Command

Run the following safe command:

```
Invoke-AtomicTest T1059.001 -ShowDetailsBrief
```

What the Command Does

This command performs the following actions:

Step	Action
1	Installs the Invoke-AtomicRedTeam PowerShell module
2	Imports the module into PowerShell
3	Displays brief details for the T1059.001 Atomic Red Team test
4	Generates PowerShell and Sysmon process telemetry
5	Allows Wazuh to detect Atomic Red Team framework activity
6	Triggers the custom Atomic Red Team detection rule

The `-ShowDetailsBrief` option is used to keep the test controlled. The purpose is to detect framework activity and technique-related command usage, not to execute destructive payloads.

Expected Telemetry

The expected telemetry should show:

Expected Field	Expected Value / Pattern
Process	<code>powershell.exe</code>
Command Line	Contains <code>Invoke-AtomicTest</code>
Technique	<code>T1059.001</code>

Event Source	Sysmon Event ID 1
MITRE Technique	T1059.001
Wazuh Custom Rule	100109

Wazuh Dashboard Search

After running the command, go to Wazuh Dashboard and search:

rule.id:100109

If the custom rule does not appear immediately, search for:

Invoke-AtomicTest

or:

AtomicRedTeam

or:

T1059.001

or:

Sysmon Event ID 1

Expected Detection Result

The expected result is:

Item	Expected Result
Sysmon	Captures PowerShell process activity
Wazuh Alert	Generated
Custom Rule	100109
Detection Status	Detected

MITRE Mapping

T1059.001

Evidence
Screenshot

Captured from Wazuh Dashboard

Evidence to Capture

The following evidence should be captured:

1. Screenshot of Atomic Red Team command execution.
2. Wazuh alert screenshot showing rule **100109**.
3. JSON view screenshot showing command-line details.
4. Sysmon Event ID 1 details if visible.
5. MITRE ATT&CK mapping screenshot if available.

Recommended evidence filenames:

14-atomic-red-team-alert.png
14-atomic-red-team-json.png

Cleanup Required

No destructive payload was executed. Cleanup is optional.

If desired, the Atomic Red Team module can be removed with:

```
Uninstall-Module -Name invoke-atomicredteam -AllVersions  
Uninstall-Module -Name powershell-yaml -AllVersions
```

If PowerShell asks for confirmation, type:

Y

Safety Notes

Atomic Red Team should only be used in authorized environments. In this project, it was used only for controlled detection validation. The command used was focused on framework activity and technique details, not destructive execution.

5. Post-Test Validation Checklist

After each detection test, the following checklist should be completed:

Validation Item	Completed
Test command executed in Windows 10 VM	Yes / No
Wazuh received the event	Yes / No
Custom rule fired	Yes / No
MITRE mapping visible or documented	Yes / No
Alert screenshot captured	Yes / No
JSON details reviewed	Yes / No
Evidence sanitized	Yes / No
Cleanup completed where required	Yes / No
Validation matrix updated	Yes / No
False-positive notes updated	Yes / No

This checklist helps ensure every detection is complete and evidence-backed.

6. Wazuh Search Reference

The following Wazuh Dashboard searches were used or recommended during validation:

Detection	Search Query
DET-001	<code>rule.id:100 100</code>
DET-002	<code>rule.id:100 101</code>
DET-003	<code>rule.id:100 102</code>

DET-009 `rule.id:100`
 `108`

DET-010 `rule.id:100`
 `109`

Additional fallback searches:

Detection	Fallback Search Terms
DET-001	<code>powershell, EncodedCommand, T1059.001</code>
DET-002	<code>lab_t1136_test, user account created, T1136.001</code>
DET-003	<code>WazuhLabSvc, sc.exe, T1543.003</code>
DET-009	<code>EICAR, Windows Defender, T1204.002</code>
DET-010	<code>Invoke-AtomicTest, AtomicRedTeam, T1059.001</code>

7. Cleanup Summary

The following cleanup actions should be performed after testing:

Detection	Cleanup Required	Cleanup Command
DET-001	No	Not required
DET-002	Yes	<code>Remove-LocalUser -Name "lab_t1136_test"</code>
DET-003	Yes	<code>sc.exe delete WazuhLabSvc</code>
DET-009	Optional	<code>Remove-Item "C:\Users\Public\WazuhMalwareSim" -Recurse -Force</code>
DET-010	Optional	<code>Uninstall-Module -Name invoke-atomicredteam -AllVersions</code>

Cleanup helps keep the lab stable and prevents test artifacts from affecting future validation.

8. Safety and Authorization Notes

All commands in this document were designed for a controlled local lab environment. The commands should not be executed on production systems or third-party systems without authorization.

The following activities were intentionally excluded from this project:

- Real malware execution
- Credential theft
- LSASS dumping
- Ransomware simulation
- Command-and-control payloads
- Public IP testing
- Unauthorized scanning
- Third-party system testing

This controlled approach allows detection engineering practice without creating unnecessary risk.

9. Lessons Learned from Test Execution

Several important lessons were learned from executing and validating these commands.

First, detection engineering depends heavily on telemetry quality. If Sysmon, Windows Security logs, or Defender Operational logs are not being forwarded correctly, the detection may not fire even if the rule logic is correct.

Second, safe simulation is enough to prove detection visibility. Commands such as encoded `whoami`, temporary local user creation, temporary service creation, EICAR, and Atomic Red Team framework activity can generate useful security telemetry without executing harmful payloads.

Third, cleanup is part of responsible testing. Temporary users, services, and files should be removed after validation so the lab remains clean.

Fourth, Wazuh dashboard evidence is important. A detection is more credible when supported by screenshots, JSON fields, MITRE mapping, and validation notes.

Finally, every detection should be documented with its test command, expected telemetry, rule ID, MITRE mapping, evidence, and cleanup process. This makes the project easier to review, explain, and defend during interviews.

10. Final Summary

This document lists the controlled commands used to generate telemetry for the Windows Detection Engineering Lab. The commands were designed to safely trigger five validated Wazuh detections mapped to MITRE ATT&CK.

The validated command coverage includes:

Detection ID	Behavior Tested
DET-001	Encoded PowerShell execution
DET-002	Local user account creation
DET-003	Windows service creation
DET-009	EICAR Defender malware-test detection
DET-010	Atomic Red Team framework activity